JACOB VALENZUELA

Software Engineer • Delivery & Flow Analytics
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SUMMARY

Software engineer focused on team effectiveness, delivery discipline, and flow analytics. On a 20-engineer team building a mission-critical internal application, I partner with our Product Owner on planning, sizing, and process, and built the team's first flow analytics dashboards in Azure DevOps — WIP aging, PR aging, and flow-metric estimation — giving leadership and the PO visibility we previously lacked. Also drove team-wide adoption of automated testing standards and mentored engineers on practice. Believe data analytics are essential to modern software delivery and that as AI absorbs more of the technical work, ownership, project management, and team effectiveness become the higher-leverage skills.

SKILLS

Delivery & Team Effectiveness: Product Owner partnership, sprint planning, backlog grooming, ticket sizing, WIP limits, flow metrics, delivery analytics, Azure DevOps dashboards, mentoring, technical writing
Testing & Quality: Test strategy, automated UI/integration testing, JUnit, SonarQube, code review, CI test integration, QA enablement
Languages: TypeScript, JavaScript, Java, Python
Frontend: React, Redux, vis.js, ArcGIS Esri Maps, HTML/CSS
Backend: Spring Boot, REST APIs, OpenAPI/Swagger, Apache Kafka, Docker, Microservices
DevOps & Tooling: Git, Azure DevOps CI/CD, Maven, Gradle, Docker Compose, WSL, ESLint, Prettier
Currently exploring: LLM/agentic systems, AI-driven workflow automation

EXPERIENCE

Software Engineer — KBR (June 2022 – Present)
Mission-critical internal application serving dozens to hundreds of government users; 20-engineer team.

Testing Leadership & Mentoring

- Drove team-wide adoption of automated testing best practices on a 20-engineer team where no shared testing discipline existed; established the team's first testing playbook and standards.
- Mentored the full 20-person team on test strategy, SonarQube standards, and CI integration; elevated quality bar across the team.
- Run recurring 1:1 mentorship with 1–3 engineers on testing, refactoring, code review, and clean architecture.
- Own automated UI test infrastructure end-to-end; authored testing documentation now used by both QA and engineering.

Delivery Practice & Flow Analytics

- Partner closely with the Product Owner on scope, prioritization, and delivery trade-offs across the 20-engineer team.
- Reworked the team's ticket sizing, sprint planning, and backlog grooming practices to produce more predictable delivery.
- Introduced WIP limits and PR-aging policies that reduced stale work and surfaced flow bottlenecks.
- Built the team's first flow analytics dashboards in Azure DevOps — WIP aging, PR aging, and flow-metric estimation — giving leadership and the PO visibility we previously lacked. Team had only tracked raw throughput before this.
- Think like a product owner and engineering manager: framing technical work in terms of user outcomes, delivery risk, and team capacity, not just implementation detail.

Engineering Delivery

- Owned major frontend components end-to-end: led the refactor/rewrite of a critical ArcGIS Esri map and the design/build of a vis.js-based timeline now used as core UI surfaces. Authored the team's UI Style Guide.
- Owned multiple Spring Boot microservices end-to-end; designed and documented RESTful APIs with OpenAPI/Swagger and implemented Apache Kafka messaging between services.
- Diagnosed and resolved multiple high-impact frontend performance issues blocking user workflows; modernized a legacy Python API enabling remote browser automation.

AI & INDEPENDENT WORK

- Exploring agentic AI workflows in production at KBR and on personal projects; daily user of Claude (Max-tier subscription) for engineering and automation.
- Exploring AI-driven workflow automation for small businesses as an emerging consulting direction.

EDUCATION

California State University, San Marcos — B.S., Software Engineering (Spring 2023)